Dwight Alan Meglan, Ph.D.

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Medical Device Technology Developer & Creator · **Surgical Robotics, Simulation & Image-Guided Systems**

FOUNDER · CHIEF TECHNOLOGY OFFICER · INVENTOR · TECHNICAL DILIGENCE & EXPERT WITNESS

35+	15	90+	>$15M	6+
YEARS BUILDING MEDICAL DEVICES & SYSTEMS	COMMERCIAL SURGICAL ROBOT PROGRAMS	U.S. & INTERNATIONAL PATENT FAMILIES	RESEARCH GRANTS AUTHORED & WON	VENTURES FOUNDED MULTIPLE EXITS

SIMULATION & TRAINING

- SimQuest — CTO · surgical simulation
- mentice — CTO · VIST simulator
- immersion — HT Medical · medical VR & haptics
- Musculographics Inc. — musculoskeletal simulation
- FIRST — Team 1757 lead mentor · 2008–present

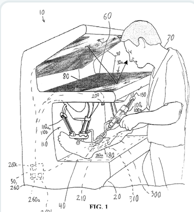
MEDICAL ROBOTICS & DEVICES

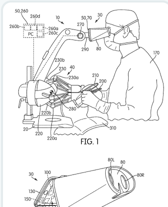
- Medtronic — Hugo · 3D-viz lead / fellow
- Johnson&Johnson — Ottava robotics
- AURIS™ — iPlatform / Monarch
- COVIDIEN — robotic surgery
- hansen MEDICAL — co-founder · architect
- OLYMPUS — Spirus spiral endoscope · $60M exit
- endoVia — first NOTES robot
- ImperativeCare — stroke robotics
- Thermalin Inc. — insulin patch pump
- HeartLander — epicardial robot · founder
- Cardathea — CSP pacing guidance · founder & CTO
- Angio8 — endovascular robot · CTO
- ISOLA THERAPEUTICS — lung therapy · CTO
- elements ENDOSCOPY — modular colonoscope · founder & CTO
- ROCKYMOUNTAIN BIPHASIC — cardiac stimulation · CTO
- exemplar devices — consultancy · development · expert witness

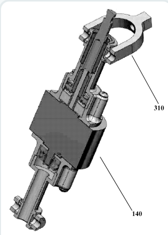
RESEARCH & INDUSTRY LABS

- MITSUBISHI ELECTRIC — Director, Medical Applications · MERL
- MAYO CLINIC — orthopedic biomechanics fellow
- The Ohio State University — Ph.D. · gait analysis laboratory
- UCC — ASSERT Centre · visiting professor
- Carnegie Mellon University — HeartLander origin license
- Foster-Miller — biomedical business development

02 Illustrated Summary: Machines & Systems from Granted Patents


US 9,607,528


US 9,092,996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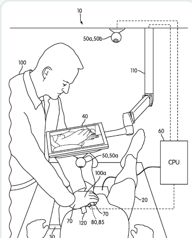
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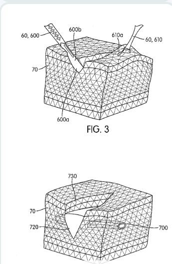
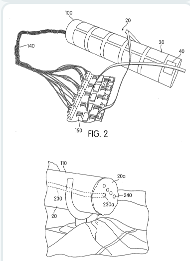

US 2010/0178644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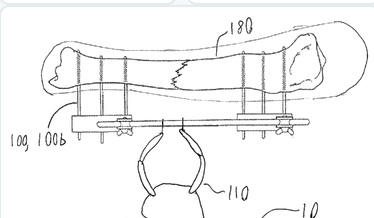
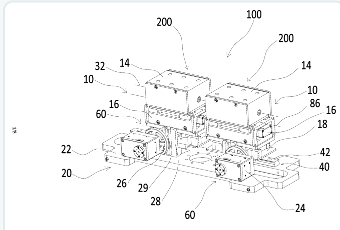
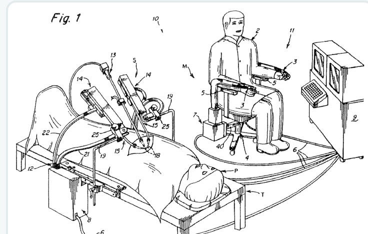
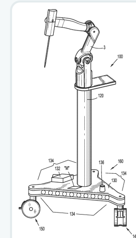
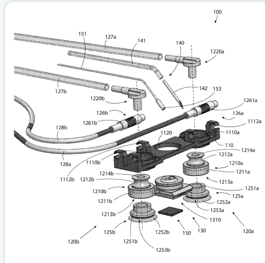

US 9,607,528 FIG. 2

SIMULATION & TRAINING


WO 2024/051854


US 7,699,835


US 11,986,261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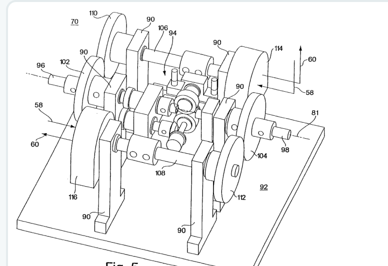
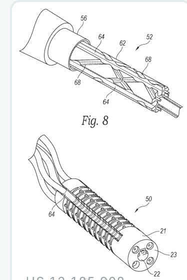
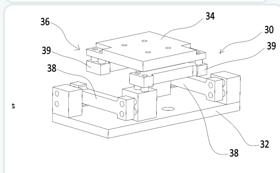
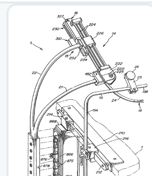
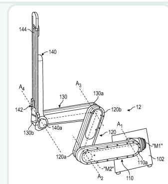


Fig. 5 US 6,096,004


US 12,185,908


WO 2024/051854


US 7,959,557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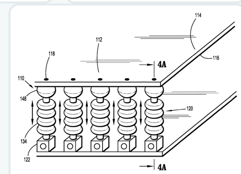
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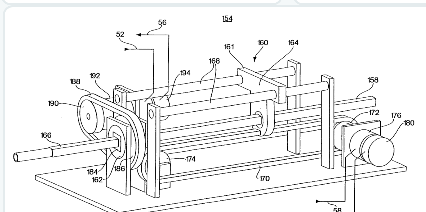


Fig. 11 US 6,096,004

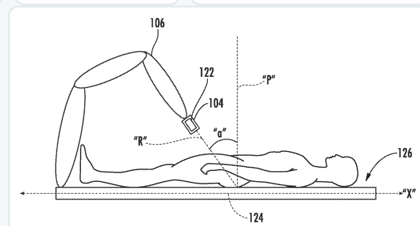
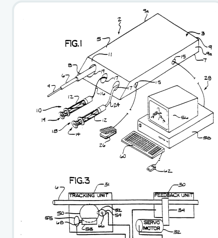
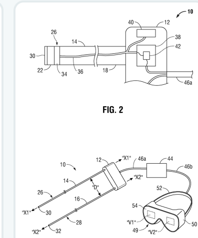
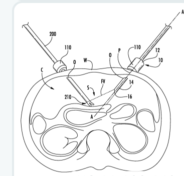


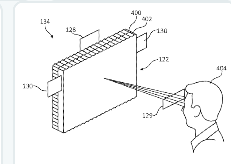
FIG. 2 US 11,553,984

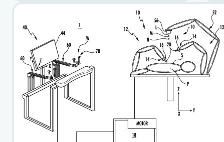
MEDICAL ROBOTICS & DEVICES

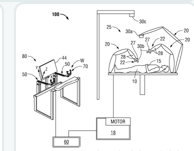

WO 1998/010387


US 11,058,288


US 11,123,149


US 11,547,520


US 10,660,714


US 10,834,332

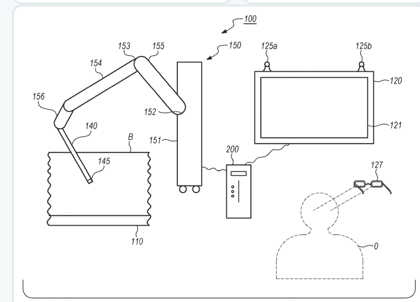


FIG. 1 US 11,647,888

VISUALIZATION, SENSING & GUIDANCE

All drawings are from **granted U.S. and international patents on which Dr. Meglan is an inventor**: public records; each figure links to its patent on Google Patents.

03 Medical Device & Robotics Experience: Role & Contribution Matrix

PERIOD	ORGANIZATION	ROLE	PROJECT / FOCUS	ARCHITECT	FOUNDER / CTO	DETAILED DESIGN	BUILD & TEST	PRODUCT / TRIAL	IP
2023–pres	**Cardathea** •	Founder & CTO	CSP pacing-lead placement guidance	✓	✓	✓	✓	✓	✓
2022–pres	**Rocky Mountain Biphasic** •	CTO	Novel cardiac / insulin stimulation	✓	✓	✓	✓	✓	✓
2022–pres	**Elements Endoscopy** •	Founder & CTO	Modular drive-by-wire colonoscope	✓	✓	✓	✓	✓	✓
2022–pres	**Isola Therapeutics** •	CTO	Aerosolized / fluidic lung therapy	✓	✓	✓	✓	✓	✓
2021–pres	**Angio8** •	Founder & CTO	Endovascular catheterization robot	✓	✓	✓	✓		✓
2017–pres	**Exemplar Devices** •	Principal / Consultant	Device development · diligence · expert witness	✓		✓	✓		✓
2019–2023	**Auris / J&J Robotics**	Consultant	Ottava robot: full-arm contact sensorization	✓		✓	✓		✓
2013–2017	**Medtronic (Covidien)**	System Engineer Fellow	Hugo robot: 3D-visualization lead	✓	✓	✓	✓	✓	✓
2013–2019	**Univ. College Cork**	Visiting Research Prof.	Surgical-training research (ASSERT)						
2005–2025	**HeartLander Surgical**	Founder & CTO	Epicardial VT-therapy robot (NIH SBIR) · Cardathea predecessor	✓	✓	✓	✓	✓	✓
2010–2011	**Spirus → Olympus**	Independent Designer	Powered spiral-endoscope electronics	✓		✓	✓	✓	
2004–2013	**SimQuest International**	CTO	Surgical simulation (>$14M in grants)	✓	✓	✓	✓	✓	✓
2003–2004	**Foster-Miller**	Biomed Bus. Dev. Lead	Medical-device practice expansion						
2002–2003	**Hansen Medical**	Chief System Architect	Telerobotic catheterization (co-founder)	✓	✓				✓
2001–2002	**endoVia Medical**	Sr. Program Manager	First endoluminal NOTES robot	✓	✓	✓	✓	✓	✓
1999–2001	**Mentice**	CTO	VIST interventional simulator	✓	✓	✓	✓	✓	✓
1996–1999	**Mitsubishi (MERL)**	Director, Medical Apps	Interventional-cardiology simulator	✓	✓	✓	✓	✓	✓
1994–1996	**HT Medical → Immersion**	Engineering Coordinator	Virtual surgical environments (VR)	✓	✓	✓	✓		✓
1993–1994	**MusculoGraphics**	Sr. Research Scientist	Musculoskeletal-simulation software	✓		✓	✓		
1991–1993	**Mayo Clinic**	Research Fellow	Orthopedic biomechanics	✓		✓	✓		
1989–1991	**Ohio State University**	Research Engineer	Gait-analysis laboratory			✓	✓		

• = active engagement. ✓ = direct, hands-on contribution in that capacity. Additional consulting / advisory: **Neptune Medical** (GI robotics) · **Imperative Care** (stroke robotics) · **Abbott** · **Thermalin** (insulin patch pump) · **Dyme Medical** (insulin delivery) · **MIIS** (imaging).

04 Intellectual Property: 90+ Patent & Application Families

Inventor on **90+ U.S. and international patent and application families** spanning surgical robotics, simulation, endoscopy, imaging, and interventional cardiology; assignees include Covidien LP (Medtronic Surgical Innovations), Cilag (J&J), Hansen Medical, Mitsubishi Electric, Immersion, SimQuest, Elements Endoscopy, and HeartLander. Foundational grants include **US 6,096,004** (master/slave control of tubular medical tools, the basis of Hansen Medical) and **US 11,547,520** (robotic device & viewer-adaptive stereoscopic display). Patent numbers below link to the full records.

PATENT	TITLE	ASSIGNEE
US 12,567,506	Simulation-based surgical procedure planning	Cilag (J&J)
US 12,491,028	Suturing guidance system	Covidien LP
WO 2025/226661	Endoscope handle sterilizer	Elements Endoscopy
US 12,440,286	Partial-reusable endo-tool robot	Foreveryoung Tech.
US 12,357,406	Robotic surgical collision detection	Covidien LP
US 12,272,463	Methods for surgical simulation	Cilag (J&J)
US 12,262,863	Image mapping & fusion during surgery	Covidien LP
US 12,185,908	Endoscope with IMU / haptic controls	Elements Endoscopy
US 12,106,508	Mitigating collision of a robotic system	Covidien LP
US 12,102,288	Multi-use endoscopes	Elements Endoscopy
US 11,986,261	Surgical robotic-cart placement	Covidien LP
WO 2024/051854	Catheter robot with force measurement	Shenzhen Tonglu
US 2024/0099767	Medical diagnosis & treatment system	HeartLander
US 11,583,356	Controller for imaging device	Covidien LP
US 11,548,140	Radio-based arm-cart location	Covidien LP

PATENT	TITLE	ASSIGNEE
US 11,547,520	Robotic device & stereoscopic display	Covidien LP
US 11,529,038	Endoscope with IMU / haptics	Elements Endoscopy
US 11,258,964	Camera-viewpoint synthesis in MIS	Covidien LP
US 11,234,883	Operating table for robotic systems	Covidien LP
US 11,123,149	Angled-endoscope visualization	Covidien LP
WO 1998/010387	Interventional-radiology interface	Immersion
US 10,834,332	Camera viewpoints in MIS	Covidien LP
US 10,660,714	Optical force sensor	Covidien LP
US 9,607,528	Combined tissue & bone simulator	SimQuest
US 9,142,144	Hemorrhage-control simulator	SimQuest
US 9,092,996	Microsurgery simulator	SimQuest
US 8,915,743	Burr-hole drilling simulator	SimQuest
US 8,671,950	Robotic medical instrument system	Hansen Medical
US 7,699,835	Robotically controlled instruments	Hansen Medical
US 6,096,004	Master/slave tubular medical tools	Mitsubishi MERL

FURTHER GRANTED & PUBLISHED FAMILIES: A SELECTION

US 12,453,548 · US 12,310,687 · US 12,256,890 · US 12,193,765 · US 12,029,517 · US 12,004,773 · US 11,576,739 · US 11,517,183 · US 10,925,636 · US 10,779,856 · US 10,251,672 · EP 3,359,075 · US 11,553,984 · US 11,058,288 · US 7,959,557 · WO 2021/242681 · WO 2021/133483 · WO 2021/067591 · WO 2019/204013 · WO 2019/204011 · US 2020/0184638 · US 2019/0088162 · US 2018/0310875 · US 2010/0178644 · US 2009/0011394 · US 2007/0123748

Plus 40+ additional U.S., European, Japanese, Chinese, and PCT filings across all domains; full list available on request, or via Google Patents.

05 Grants · Signature Achievements · Recognition

GRANTS & CONTRACTS AUTHORED: >$15M AWARDED

Precise Minimally Invasive Treatment of Ventricular Tachycardia	NIH Fast-Track SBIR
Modular Flexible Semi-Disposable Endoscope Platform	NIH SBIR I
Drug Delivery by Endoesophageal Lavage for Esophageal Cancer	NCI SBIR I
HeartLander as an Epicardial Injection-Delivery System	NIH SBIR I
Open-Source Surgical Simulation for Combat Casualty Care	U.S. Army
Intracranial Hematoma / Burr-Hole & Trauma-Flap Simulator	Army SBIR I / II / III
AR System to Teach Hemostatic-Agent Use	NIH SBIR I
Simulation-Based Wound-Closure Training System	NIH SBIR Fast-Track
Haptics-Optional Surgical Training System	Army SBIR I & II
Simulation-Based Open-Surgery Training System	Army SBIR I & II
Simulation-Based Dental Training System	NIH SBIR I & II
Leg-Trauma Wound-Simulation Trainer for Army Medics	DARPA

SIGNATURE ACHIEVEMENTS

- **$60M Olympus exit**: sole developer of the UI & motor controls for Spirus Medical's powered spiral endoscope.
- **Medtronic Hugo**: 3D-visualization lead & technical fellow; low-latency stereoscopic imaging to the da Vinci standard.
- **First NOTES surgical robot**: built the first endoluminal master–slave telerobot; assets passed to Hansen & Intuitive.
- **Mentice VIST**: designer / lead engineer of the most successful surgical-simulation product; 400+ units, 20+ years on market.
- **Co-founder, Hansen Medical**: recruited by Intuitive founder Fred Moll; foundational catheter-robot IP (US 6,096,004).
- **Medtronic robotics ML group**: identified & advocated the acquisition (Digital Surgery).
- **OpenSurgSim**: conceived & won funding for the DoD open-source surgical-simulation platform.
- **Technical & IP due diligence**: advisor to investors and acquirers of surgical-robotics and simulation companies.
- **Expert witness**: medical-device & simulation litigation; patent & trade-secret assessment, reports, depositions.

RECOGNITION & SERVICE

- Senior Member, IEEE (2009)
- Senior Member, ACM (2009)
- NIH Study-Section Reviewer
- U.S. Defense SBIR Reviewer
- Canadian NSERC Reviewer
- ACS Steering Committee on Surgical Simulation (2001)
- FIRST Robotics 1757 Lead Mentor: NE District Champions 2023 · Worlds top 4%

SELECTED PUBLICATIONS

- Coles T, Meglan D, John N. "The Role of Haptics in Medical Training Simulators." *IEEE Trans. Haptics*, 2011;4(1):51–66.
- Meglan D, et al. "Development of an Open Surgery Simulator." *Medicine Meets VR*, 2011.
- Champion HR, Meglan DA. "Minimizing surgical error by incorporating objective assessment." *J. Am. Coll. Surg.*, 2008.
- Dawson SL, et al. "Computer-based simulator for interventional-cardiology training." *Catheter. Cardiovasc. Interv.*, 2000.